

13 Tracing the music actor-network

Losing the meaning of musical experience? The limits of a routinization of science and technology studies applied to techniques and knowledges of music

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The power of music to hold out against comprehensive description is simultaneously a cliché and a glaring reality when you try to stay closest to its phenomenality. The development of science and technology studies as an international and multifaceted academic world, of sociology of translation in France (Callon and Latour 1981; Latour 1988; Akrich 1992; Callon 2001), and, concomitantly, the emerging of sociology of mediation (Hennion 1993, 2009, 2013, 2017), has more or less directly helped to take this trouble seriously. To do so, these studies had to come out of the social sciences' routines concerning the ways of thinking about technology and music. But, along the way, new own routines may have been shaped, without us truly realizing.

What can and cannot science and technology studies (STS) teach us about the experiencing of music? How does studying musical experience query STS's relevance? What I am proposing is to address these issues, focusing on the specific theme of the digital transformation of the music industry and its consequences for listening ecology and music experience. The key idea of this chapter is to show, at the same time, the contribution of STS's perspectives—especially so-called actor-network theory (ANT) (Callon and Latour 1981; Callon 2001; Latour 2005, 2012) and, to a lesser extent, infrastructure studies (Star and Ruhleder 1996; Star 1999)—to study current forms of music exchanging and the limits of a routinization of these perspectives to understand the meaning of musical experience.

To do so, I will mainly rely on two of my own investigations. The first one was a long-term ethnography of a record shop, which gradually became a site of online sales before migrating onto the Discogs platform, an opportunity to document the transition from physical trade to e-commerce of music. Directly linked to this first step, the second one was to analyze the various generations of *recommender systems* developed on the major music sites, as well as on smaller ones, in order to provide a detailed description of current forms of musical choices. A somewhat similar approach was recently conducted by other researchers, in the course of an extensive collective survey on Spotify (Eriksson et al. 2018); that will be another source of insight into a larger critical questioning. Contrary to the common misconception of the “digital revolution,” the crosscutting issue here is: what kinds of conversion, continuity, and discontinuity have occurred with the digital switchover?

Combining anthropology of knowledge (Goody 1977, 1986) and sociology of mediation (Hennion 1993, 2009; Hennion, Maisonneuve, and Gomart 2000), the ecological approach (Gibson 1979), and influences of ethnomethodology (Garfinkel 1967), this study is part of a process that shares some views with STS: (1) it is important not to confuse the outcomes of practices with their origins; (2) cultural practices are both material and social (material and semiotic), and they are extremely heterogeneous; and (3) what is called “social” is located in relations. These are obviously three cross-cutting assumptions of embedded approaches of STS, ANT, and sociology of mediation, regardless of their differences and their own pathways: STS are the broader set, including several research traditions and divergences; actor-network theory (ANT) was the name granted to the sociology of translation in French STS tradition, as soon as it was internationally recognized; from a shared journey with ANT (Hennion 2013), and especially with Bruno Latour (Hennion 2016), sociology of mediation and sociology of attachments (Hennion 1993, 2009, 2017; Hennion, Maisonneuve, and Gomart 2000) have traced their own way about musical ontologies in a pragmatist perspective (Hennion 2013, 2016). The kind of problematization that supplies my ethnomethodology-inspired work, has moved away from sociology of translation, while it remained close to Hennion's pragmatism.

In proceeding with this inquiry, I have borne in mind this principle so integral to actor-network sociology: diverse types of “translation” are needed to “embed” practices in new technology—this influence was indeed tacit until I realize I was trying to analyze “black boxes” of musical recommendation systems. The “principle of generalized symmetry” is certainly a good guide to explore the “black boxes” of digital infrastructures. But, as we will see, this

investigation has also shown that those black boxes make both less and more than “translation.” Most significantly, delving into the black boxes of digital artifacts and data may separate us from current musical experience. That’s why we should also get out of the digital technoscience way of living to clarify the limits of some of STS’s routines applied to techniques and knowledges of music.

In the first section of this chapter, I will present the transition from “physical trade” to “e-commerce” of music. I will then focus on the development of streaming platforms and of recommender systems. To conclude, I shall come back to musical experience.

From record shop to website: the first step of digital transition

My investigation began in the early 2000s in a new record shop located in the city of Lille, north of France. This independent record shop was mostly dedicated to British styles of that time: drum ‘n’ bass, 2-Step, UK Garage, Dubstep, etc. Yes, it was a scarce specialization and a niche market in France. But, in my observation over more than 10 years in different kinds of record shops in France and England, what was going on there did not really depend on either the store format or the musical styles.

What was going on cannot be reduced solely to the market dimension of exchanges. A record shop emerges as a musical place from different practices—listening, record uses, gestures, conversations, shared attention, etc. Musical things, musical skills, and culture—music itself—as well as music lovers and the expert record dealer, as such, also emerge from these common and usual practices. Material record bins and record classifications design music scapes. They are based on categories and knowledge registers produced throughout the history of recorded music (musical genres, ages, tones, etc.), which then become potential affordances to choose a record, to comment on it, even to get someone else to listen to one’s own listening (Debruyne 2015, 2016). The shared listening is linked up with listenings performed in other places (at home, in concerts, online), as well as with those of experts (the music press or amateur blogs). The record store place is a whole region of production of a musical culture, which is at least a culture of fitted listening.

Of course, a record store is a kind of sociotechnical apparatus, a kind of *sociotechnical network*, in the language of STS. Of course, taking seriously the material aspects of the place, especially the records as stakeholders and as sociotechnical things, was a good habit to describe what was going on. But while I was examining events there through this perspective, I was getting off the meaning and part of the social texture of practices I observed. I will come back to that in the last section.

In 2004, the record shop in Lille also became an online store. Its design was widely based on practical know-how crafted through the shop experience. Musical categories and record classifications all appeared on the website, which gave an overview on what was the material store layout, while making explicit (ex-)tacit categories—and, in particular, probably, the most important one: “news.” The structure of the website was very simple: it replicated the possible paths in the record bins. But its *modus operandi* was quite different: it was entirely organized under the “logic of writing” (Goody 1986) and under the mode of screen use. Like many websites of the early 2000s, it integrated a *content-based* recommender system. The recommendations were then produced from musical similarities, which were themselves produced by specialists who characterized the musical properties of each track. The calculation algorithm then allowed for recommendations of the most “similar” songs. On the store’s website, in addition to data associated systematically with the tracks (artist name, title, musical style, record label, record format), the similarities were calculated from annotations made by the music shop manager himself. So the recommender system involved the manager’s skills on the one hand, to value and to classify the music. But it did so without submitting his valuation to any form of conversation or acknowledgment that often occurred in the store. The preexisting categories and registers were, on the other hand, embedded in the algorithm calculation. But they were converted into digital equivalent data. In other words, the music valuation and expertise that prevailed in the record shop moved to the computing side. The production of discrete units required a selection from among the music genre categories and from among personal and social ways of characterizing the music. This selection was no longer subject to situated and contingent interactions. The computing side required a conversion from heterogeneous forms of music knowledge into homogeneous data. This first step of digital transition was, therefore, also a first uncoupling from music experience, even if it had always been rooted in it.

The physical store closed in 2007, the website in 2010, both for economic reasons. In 2012, the record shop resurfaced on the Discogs platform. Discogs was first a huge recording database, dedicated in the early years to vinyl records of electronic music, then broadening to every kind of recording in every style. It then also set up an ordering section, which links many buyers to many sellers who are mostly experts in their musical spheres (including a number of record stores). Discogs embeds a recommender system, but its computation database is composed of backgrounders, which are the joint basic material of every platform section. Those backgrounders look like the kind of thing a record shop manager could produce. But they are crafted by every kind of user, including many users who only use the platform as a musical knowledge database. The backgrounders can only be edited if they are collaboratively endorsed by other users. They are highly accurate, combining usual information (artist name, title, record label, publishing year, etc.) and a heap of information concerning the materiality of the record and its biography (original release/reissue, years of publishing, country of publishing, players on each track, record format, background of record cover, producers, recording studio, etc.). On the commercial section, sellers add an assessment of the material quality that is based on a common evaluation grid. This assessment and its seller are themselves evaluated by buyers. Discogs is a place where physically missing records are constantly “rematerialized:” the material quality seems to be a broader problem now that music mainly travels through digital formats. On Discogs, of course, this tendency relies on its records’ amateur background. It requires an ongoing collective inquiry and involves the question of trust. These inquiries seek to test the data validity; they are grounded in trust about users’ cross-expertise. But the trust is never permanently anchored: it is constantly valuated and based on distrust about musical data flowing on the internet.

To follow that record store’s life also means following one possible way in which music experience might change. On the one hand, digital infrastructure “embeds” a common cultural background and is grounded in users’ know-how. On the other hand, the process through which it does so converts heterogeneous and contingent forms into homogeneous data. In other words, from record shop to e-commerce of music, musical skills (and culture) are involved in digital exchanges, but musical skills (and culture) shaping is separated from its social and material situation. The massive way of digitalization relies on this first step, but may be quite different in regard to music experiencing. Let’s explore it.

Recommender systems and streaming music: testing STS’s routines

The 2010s were the time of the parallel development of (a) so-called “social networks,” (b) recommender systems, and (c) streaming music platforms.

The development of recommender systems linked to digital forms of music exchanges has progressively settled new recommendation formats. They have become increasingly sophisticated, eased by a huge computational power; they are underpinned by endeavors to encode every practice and every moment of music experience; they are also informed by self-reliant exchanges within digital and connected device spaces.

I shall not go into the details of this sophistication. Rather, I wish to single out three main issues:

- (1) the recommender systems hybridize the “item-centric” recommendation—close to the one I described within the online shop but using automatically generated “content-based” data, “user-centric” recommendation—that first characterized commercial platforms and navigation systems, “collaborative filtering” recommendation, inspired by i-Tunes and social network habits, and sometimes, experts’ playlists;
- (2) they combine data within the platform concerned and data from many other connected spaces—by conducting extensive acoustic and textual analysis, in order to shape nearness for items and for users (“nearest neighbor” analysis); and
- (3) they are much more “personalized”—by raising the profile of each user and by taking account of the user’s daily listening, in order to come up with everyday life recommendations.

We can say that those systems build on a set of sameness: sameness of users’ taste, sameness—or musical identicalness—of songs, sameness of semantic universes used to talk about music. They also build on a regularity of our practices and our taste. So they are combinatory systems of the greatest degree of sameness and of the strongest regularities.

But, in fact, the algorithms create these samenesses and regularities: (a) they encode heterogeneous things into consistent digital data; (b) “nearest neighbors” are implemented within this data infrastructure and through a continuous data mining process. It is no doubt true that we tend to make regular choices that correspond with our own taste, but these systems transform contingent regularities into mathematical rules, so that the recommender systems work from a complete disconnection of music experience, understood as the meaning produced in contingent and heterogeneous situations of listening.

In some ways, recommender systems are quite “efficient:” we are quite satisfied with their recommendations, including the discovery of new bands. This relative “efficiency” depends arguably on the database used and on the relevance of its samenesses. But data are not exactly an embedded cultural background: they transfigure it into general equivalence detached from musical experience. Furthermore, the internal calculation of similarities and of regularities enhances the congruence between possible uses so that the “efficiency” is matched by conformism potential, which seems to be mathematically irremediable. But it could mean general conformity in musical taste only on the assumption that websites would be the exclusive means of access to music. This is obviously not the case, because many of us still share records with friends, read papers, go to a concert, or dance to music we will never log in out streaming playlist. Even so, the internet is more than ever the preferred site for music lovers, who are able to combine different musical devices to prevent conformist choices. It is also a great space for casual users, who only use streaming platforms and who enjoy “helpful” recommender systems.

Comparison of what’s happening on Discogs with uses of major streaming platforms could appear as highly indicative of this growing gap between casual listeners and amateur experts. But, on the one hand, music lovers could use both types of platforms. On the other hand, the comparison shows that the use of similar recommendation systems doesn’t necessarily accompany the same kind of experience. I will return to that point later.

To undertake my investigation, I have linked a long-term ethnography of a record store offline and on—as discussed in the previous section—to an analysis of data science literature and of streaming music interfaces, and to an inquiry into uses of them. It was a way to go into the “black boxes” of digital recommendation systems, to understand their *modus operandi* and to compare them to what was going on in record shops. As Bruno Latour (2012) mentioned, the mantra of efficiency—especially what he called “double click fantasy”—should not be confused with our own questions. In the case of a recommender system, the efficiency fantasy is part of the system itself, which is also a belief system.

But at the same time, the question of modes of being often shifts towards the question of successes and failures. That’s one of the risks of STS’s routines: to fall back into causal explanation, though it is more interesting to know under what conditions a causal explanation is relevant in practice. Another risk is to make “black boxes” the only reason for investigation: does this reflex of investigating black boxes really help us to understand the meaning of what’s going on?

Trying to answer this question, it could be interesting to briefly review a fascinating collective inquiry on Spotify by Eriksson et al. (2018), which is, in some ways, complementary to mine and reaches similar conclusions. This study is grounded in two different STS traditions—ANT and infrastructure studies (Star and Ruhleder 1996; Star 1999)—and was conducted in part by adopting these STS reflexes I propose to question. Rather than summarizing it, I focus on what is close to the path followed in my own investigation.

The authors tell the story of Spotify by following the stages, and by tracing “the manifold transactions” (Eriksson et al. 2018: 6), from small independent enterprise to major corporation of digital music, including experimental techniques to enter Spotify’s “black box.” In doing so, they follow “the process by which Spotify created lasting asymmetries between itself and us, in terms of capital as much as cultural power” (Eriksson et al. 2018: 7). The major find is that they tell this story by focusing on the financialization of the system as a crucial dimension of each cultural and technological stage. The most important step is undoubtedly what they call the “curatorial turn.” It occurs when, from 2014, Spotify enters the competitive world of recommendation and fairly quickly becomes its most influential competitor. This “curatorial turn” is equally economical, technological, and cultural. It involves developing sophisticated recommender systems in order to offer personalized recommendations and spreading a rhetoric of unique music “experience.” Then, reflecting on the issues of how files become music and how Spotify packages music, they point to three research outputs, among others, which substantiate my argument. The first one is that algorithms enable regularity. The second one is that digital infrastructuring disconnects streaming experience from “the singularity of the music experience” (Eriksson et al. 2018: 80). The third one is complementary to the idea of a growing gap between casual listeners and amateur experts: “The emphasis of delivering recommendations in the Spotify interface forges an asymmetrical relation between casual users and musical experts” (Eriksson et al. 2018: 120).

The book title itself—*Spotify Teardown*—makes clear that we are entering the field of STS inquiries, here particularly inspired both by Michel Callon and Bruno Latour’s ANT theory, and infrastructure studies in the line of Susan Leigh Star. It is necessary here to mention a few limitations of this book, which seem to me to be broader limitations of STS perspectives, regardless of the different variations and discrepancies between ANT and

Infrastructure Studies. The methodology's inventiveness is one of the undeniable qualities in this book. To enter the Spotify "black box," researchers have employed experimental back-end methods such as bots, or small software scripts, in order to formulate assumptions concerning streaming infrastructure. In doing so, they likely suggest valid assumptions. But they do not really enter the black boxes; they do not really describe the logic of calculation itself. This logic could, however, be understood by analyzing data science literature, which has been available on the internet for years now. It doesn't require cutting-edge mathematical skills, as illustrated in my description above. This is a recurring limitation in ANT tradition: the metaphor of the "black box" seems to be sufficient unto itself and the investigation's focus on upstream and downstream seems to be appropriate for describing reality, so that the very logic, the very *modus operandi* of digital infrastructure, is not questioned. Conceiving Spotify as an infrastructure of interwoven global networks and data misses what this logic produces. Computer operations require and produce a separation between material infrastructure, which locates the results of this invisible work in our physical world (the world of experience), and symbolic mathematical infrastructure, which could be supported by any other material infrastructure: this distinction between *hardware* and *software* bases the very possibility of computing. To use the words of Susan Leigh Star (1999), data and their ecology are neither *artifacts*, nor *traces* nor *representations*: they proceed in a world which is not accessible in experience space-time (even if, of course, they are conceived upstream as traces and records, used downstream as artifacts, and, on both sides, full of representations). That's probably why "experience" is not really problematized in *Spotify Teardown*: it could finally be understood as Spotify's meaning of "experience" (which is certainly not the authors' intention). Before taking up this question more broadly, I may suggest introducing what Harold Garfinkel has called "background expectancies" (Garfinkel 1967). Everyday life rules proposed by Spotify are based on things which are simply not "factual" in everyday experience: users do not achieve these rules by using Spotify applications, because there is no "background expectancy" shared with artifacts. In the same vein, digital data do not perform any identity, because deep learning systems do not need categorical types (or sense in fact) to aggregate data, to get them working together, and to organize recommendations, contrary to statistical surveys. In some ways, these limitations coincide with some of STS's limitations for thinking about experiencing.

Back to the question of (the meaning of) musical experience

The aim here is definitely not to go back to the idea of something like a primary experience or to a conception of a musical experience as a pure/purified one. In line with the philosophy of John Dewey (1905, 1934, 1977), pioneer of STS in many ways, the experience we make or the experience we have results from our transactions with our environment. An experience consists of actions and their consequences. We cannot understand actions without taking their consequences into account. Those consequences don't mean anything without taking actions into account. Experience also weaves continuity between past and present. It is always situated, active, and unified. Against dualism of body and mind, experience from Dewey's point of view involves sensitive and intelligible perception. From this point of view, musical experience results from listening practices situated in sociomaterial environments, links present to past experiences, and produces meaning in an unpredictable manner.

On streaming platforms, situated listening is not directly involved: it is mobilized through its traces and the production of similarities, their "translation" by the equity method. The calculation systems that organize their symbolic infrastructure are combinatory systems of the greatest degree of sameness and of the strongest regularities. Of course, listening combines different elements from the past and from the environment, different dimensions of our relationship with the music, various habits and cultural references. However, music experience doesn't proceed from a combinatory logic, but rather from located contingency, in which interactions and relationships interweave and in which rational activities are fewer than appropriate involvements. But, as stated above, the algorithms work in a kind of symbolic area which is inaccessible to human experience.

So digital transitions are new forms of "domestication" (Goody 1977) of music experience, on both sides: material and symbolic ones. The musical experience had been "domesticated" several times, and especially through its "record-morphic" mediation (Hennion, Maisonneuve, and Gomart 2000; Maisonneuve 2009) throughout the 20th century: this was a major stage to enhance music experience and share a flourishing musical knowledge. Regarding what's going on in experience space-time, digital domestication is very close to what Jack Goody described about transition from oral

to written technology of mind. On the one hand, digital infrastructure produces visual scapes, in which listings have appeared to be practical shapes to music data filtering. On the other hand, those universes now act (relatively) independently of other social activities.

From this point of view, what could we infer from investigations into black boxes of digital music relating to the meaning of music experience? From the 2010s, the algorithm activities have been subject neither to the intersubjective fitting required for trust nor to the authority of music experts. But, in fact, neither serious nor casual music lovers really need to “trust” recommendation systems. The more these systems become autonomous, the more they are used just for the sake of random playing and of serendipity. These ways of experiencing streaming interfaces are at least as common as rational uses to test their efficiency. The meaning of this kind of experience seems far from the unique “experience” designed by data engineering and by web-marketing. The uniqueness of our situated wanderings precisely disappears in a predictive calculation: discovering a new band by accident in a festival, dancing to a music we do not usually listen to, trying to listen to reggae because our romantic partner swears by this style...Our musical experience has the flavor of its moments and here, too, is unpredictable.

The gap between the meaning of “experience” designed by streaming infrastructures and the meaning produced by located living musical experience makes the limits of ANT’s routines more apparent. Considering the “actor-network” as something subsuming the “social” includes the risk of confusing traces of practices with the practices themselves, as well as the risk of making the regularity of uses a defining structure. For example, the hypothesis of a growing gap between casual listeners and amateur experts is probably relevant, but is not enough to understand the ways of making sense through musical experience today. Delving into black boxes could help us to understand their *modus operandi*, both social and technical, but it can’t help us to reach the experience, as the meaning stems from transactions with environments. Focusing on the homogenizing process should not lead us to forget that the meaning of experience, which is still the experience of heterogeneity, is not just one element among many others; it is what holds experience together as one. *Mediations*, *translations*, and *modes of being* include not only relating networks, but also meaning mediation (such as trust, pleasure, sharing, etc.). And this kind of mediation is what gives consistency to experience.

In my own investigations, I may have spent too much time delving into the black boxes of recommender systems. I considered, for some time, that computing techniques were efficient and succeeded in domesticating musical experience while establishing two worlds of listening (expert versus casual). I forgot relatively simple things: real practices are still heterogeneous, and the meaning of music experience is still unforeseeable. In fact, the more we focus on networks of relations and digital black boxes, the less we keep close to musical phenomena—also understood as material, social, and situated.

Now, my questioning could be stated as follows: what kind of skilled “contact” (Crawford 2015) with music, with the world, with oneself, and with the others, is required and prefigured by digital ecology? The question implies that the answer lies not only within digital infrastructures. The recommendation technology could require and prefigure an emerging shared culture, a heterogeneous experience; it could thus favor using configurations which make users more capable (full contact skills). The same technology could only apply for internal digital calculation, residual traces of cultural background, and require no other skill than (growing and continuing where possible) uses of systems themselves.

Tracing the music actor-network: disconnecting experience?

Methodological inventiveness is obviously a shared quality in science and technology studies, perhaps partly because exploring “black boxes” often requires it. But, ironically, this inventiveness often relies on a routinization of STS theoretical (and metaphorical) reflexes, here applied to techniques and knowledge of music. This chapter has sought to consider at the same time those two aspects, with emphasis on this question: what kinds of conversion, continuity, and discontinuity have occurred with the digital switchover?

Exercising together investigations on musical experience and STS may reveal some of STS’s limits. In our keenness to enter the black boxes, we sometimes come either to endorse web-marketing and web-engineering with blissful zeal or, conversely, to see a computer system only as an efficient technology to domesticate musical experience. But this means not going to the heart of the matter: the unpredictability and the meaning of musical experience as a whole.

Exploring “black boxes” of recommender systems (of streaming music) could help to understand the combinatory logic that conducts online music exchanges and to see that algorithms “embed” a part of a common cultural

background. On the one hand, this explains their efficiency at least as much as their technical sophistication. On the other hand, this efficiency heightens the risk of conformity among the music preferences. But, that conformist potential becomes effective only in the hypothesis of a musical experience confined in main to digital devices, which is really not always the case. It's right to see the digital switchover as a reconfiguration of heterogeneity into sameness. But the recommendation systems don't really "embed" a cultural background: they disconnect semiotic marks from social contingency and from material affordances. To a certain extent, applying ANT to techniques and knowledges of music produces the same kind of disconnection.

What I have proposed is to lay stress on the meaning of experience: we do not experience algorithms or data aggregations; we use screen artifacts within a broader diverse experience. So the same kind of digital engineering doesn't provide the same kind of experiencing: it depends on the collective experience upon which it was relying and on a contingency that could not be preengineered—it particularly depends on "background expectancies." The meaning of musical experience is always located, and the algorithms are definitely not its location.

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